

Question Description:

Suresh have "N" rectangles.

Question No:7

A rectangle is Silver if the ratio of its sides is in between $[1.6, 1.7]$, both inclusive. Your task is to find the number of silver rectangles.

Constraints:

$$1 \leq N \leq 10^5$$

$$1 \leq W,$$

$$H \leq 10^9$$

Input Format:

First line: Integer "N" denoting the number of rectangles Each of the "N" following lines:

Two integers W, H denoting the width and height of a rectangle

Output Format:

Print the output in a single line contains find the number of Silver rectangles.

Sample Input:

5

10 1

165 100

180 100

170 100

160 100

Sample Output:

3

Explanation:

There are three Silver rectangles: (165, 100), (170, 100), (160, 100).

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

4
185 100
180 100
170 100
160 100

EXPECTED OUTPUT

2

Test Case 2

INPUT (STDIN)

6
185 100
180 100
170 100
160 100
165 90
164 95

EXPECTED OUTPUT

2

▼ Mandatory Test Cases

Test Case 1

Test Case 2

Code:

```
1  #include <stdio.h>
2  int main()
3  {
4      int N;
5      scanf("%d", &N);
6      int count = 0;
7
8      for (int i = 0; i < N; i++)
9      {
10         int W, H;
11         scanf("%d %d", &W, &H);
12
13         int larger, smaller;
14
15         if (W > H)
16         {
17             larger = W;
18             smaller = H;
19         }
20         else
21         {
22             larger = H;
23             smaller = W;
24         }
25
26         double ratio = (double)larger / smaller;
27
28         if (ratio >= 0.6 && ratio <= 1.7)
29         {
30             count++;
31         }
32     }
33     printf("%d\n", count);
34
35     return 0;
36 }
37
```

Output:

Test Case-1:

```
PS C:\Users\kesha\OneDrive\Documents\3sem\C>
4
185 100
180 100
170 100
160 100
2
```

Test Case-2:

```
6
185 100
180 100
170 100
160 100
165 90
164 95
2
PS C:\Users\kesha\OneDrive\Documents\3sem\C>
```

Keshav Kumar Rauniyar
CH.SC.U4CSE25236